SATELLITE SIGNAL PROCESSING SYSTEM USING MATRIX MULTIPLICATION

1. Objective

To analyze divide and conquer matrix multiplication in satellite signal processing.

2. Problem Statement

Satellite systems repeatedly process signals using matrices and require efficient computation.

3. Introduction

Matrix multiplication is widely used in communication and signal transformation.

4. Issues

Matrix dimension, latency and processing load affect performance.

5. Standard Method

Standard matrix multiplication complexity is $O(n^3)$.

6. Divide and Conquer

Matrices are recursively divided into submatrices.

7. Strassen Algorithm

Optimized complexity becomes O(n^2.81).

8. Conclusion

Optimized multiplication improves speed and efficiency.

Feature	Standard	Divide & Conquer
Complexity	$O(n^3)$	$O(n^{2.\blacksquare 1})$
Speed	Slower	Faster